



BMG5109 Data Science for Engineers

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Data Science

From Big Data to Data Science

- We are living in the big data era where enormous amount of data have been continually generated, each day at increasing scales. We are collecting data all the time from everywhere and we should use them properly to generate insights and add business value.
- This exponential growth of potentially valuable “data acquisition”, compounded by the Internet, social media, cloud computing, variety of sensors and new types of mobile devices, is often referred to as big data. [1]
- Big data processing represent one of the new engineering challenge and innovation.





Data Science

From Big Data to Data Science

- Dealing with this vast amount of data would impose a great challenges, hence, novel approaches and techniques are required to address these big data problems.
- Although it seems that big data makes it possible to find more useful, actionable information,... **the truth is that more data do not necessarily mean better analyses and more informative conclusions unless combined with the right techniques and technologies.**
- Data are raw facts, information is processed data with context, and knowledge is the understanding and experience that allows us to apply information for decision-making
 - Therefore, designing and deploying a system that rely on the big data process to extract some insight for the decision making or other purposes is not a trivial or straightforward task.





Data Science

From Big Data to Data Science

- Therefore, a new field is emerging that is Data Science.
- Decision-makers of all kinds, from company executives and government agencies to researchers and scientists, would like to base their decisions and actions on **the available (accessible), reliable, complete and correct data**.
- In response to these multidisciplinary requests, a new discipline of data science is forming.
- Data scientists are professionals who are trying to gain knowledge and awareness of something not known before about data. [1]
 - **They need business knowledge; they need to know how to deploy new technology; they have to understand data processing techniques, database and warehousing, statistical techniques, most of AI techniques such as machine learning... , and data visualization design ; and they need to know how to interpret and present the results.....**





Data Science

From Big Data to Data Science

- The name of data science seems to connect most strongly with areas such as databases and computer science in general, and more specific it is based on machine learning and statistics.
 - But many different kind of skills are necessary for the profile, and many other disciplines are involved:
 - skillful in communication with data users; understanding the big picture of a complex system described by data;
 - analyzing business aspects of big data application & BI;
 - performing data mining to extract value and insight from data, knowing how to transform, visualize, interpret, and summarize big data;
 - maintaining the quality of data;
 - and taking care of security, privacy, and legal aspect of data.





Data Science

From Big Data to Data Science

- Of course, there are very small number of experts who are good in all these skills,
- and therefore, we have always to make emphasis on the importance of multidisciplinary teamwork in big data environments.....





Data Science: Let us set the framework...

- **Data Acquisition** : Sensors, Devices, DBs interfaces, Web Applications, Mobile apps, Social Media Etc. , ...
- **Data Transfer and Storage**: Consistent Data Flow, Automated transfer , PIPELINES (ETL) , Infrastructure, Structured and Unstructured Data Storage, Data Warehousing,
- **Data Preparation and Preprocessing** : Cleaning, Anomaly Detection, Data Editing and Structuring.... PREP,
- **Data Consolidating and Discovery** : Aggregation, Segmentation, Analytics, Metrics, Features extraction, Data Visualization
- **Data Learning** : Testing, Experimentations, Simple ML Algorithms, ML, Deep Learning, NLP, LLM other “AI” techniques,

At the end we need some innovation in AI Driven db solutions.





Data Science

FROM Data Collection to Data Preparation....

- Continues collection of digital data & information should lead us to the next stage as the problem of data overload becomes more intense (size and verities)
 - **However, our ability to analyze and understand big data is far behind our ability to gather and store the data.**
- Recent advances in computing power, communications, and digital storage technologies, together with the development of data-acquisition technologies, have made it possible to acquire, process and store incredible volumes of data.
- Data from the neighborhood store's checkout register, your bank's credit card authorization device, records in your doctor's office, patterns in your telephone calls, and many more applications generate streams of digital records archived in huge business databases. [1]





Data Science

FROM Data Collection to Data Preparation

- Complex distributed computer systems, communication networks, and power systems, for example, are equipped with sensors and measurement devices that gather and store a variety of data for use in monitoring, controlling, and improving their operations.
- Scientists are at the higher end of today's data-collection machinery, using data from different sources such ; from remote sensing platforms to microscope probing of cell details.
- Scientific instruments can easily generate terabytes of data in a short period of time and store them in the computer.
- One example is the hundreds of terabytes of DNA, protein-sequence, and gene expression data that biological science researchers have gathered at steadily increasing rates.
 - The information age, with the expansion of the Internet, has caused an exponential growth in information sources and also in information storage units. [1]





Data Science : Data preparation "is at the heart of ML." Felix Wick

➤ Data Preparation!!!

- Data preparation is the process of collecting, Integrating, and structuring data to be utilized in the next stages in many different ways for further applications; such as business intelligence (BI), analytics applications, data visualization applications, ML, DL, and more.....
- Activities of data preparation could include but not limited to data profiling, cleansing, validation, transformation, and storage of data both internally and externally.





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➤ why data preparation?

- Data preparation is to ensure the accuracy and consistency of the raw data and to enable it for next stage of analysis.....
- Any application that can be build on accurate, consistent, and complete data will be valid otherwise it will not be reliable.
- Data is commonly created with many types of different errors such as missing values, inaccuracies Need to be fixed.
- Data coming from different sources often have different formats that need to be combined, integrated and the formats should be unified.
- Correcting data errors, validating data and its quality and integrating data sets are big parts of data preparation objectives.





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- **Data Collection/Ingestion:** Gathering data from various sources (databases, files, APIs).
- **Data Profiling/Discovery:** Understanding the data's structure, quality, patterns, and completeness.
- **Data Cleaning (Cleansing):** Correcting errors, handling missing values (imputing or removing), and removing duplicates to improve accuracy.
- **Data Transformation:** Converting data into a suitable format, normalizing values, encoding categorical data, or applying business rules.
- **Data Enrichment:** Adding context or features from external sources (e.g., adding geolocation to addresses).
- **Data Structuring/Integration:** Combining different datasets and organizing them into a unified, usable structure (like a table).
- **Data Validation:** Checking the prepared data against quality rules for accuracy and consistency.





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- Values that Data PREP will add....
 - Data scientists usually saying that they have to allocate a lot of their time for data collection, editing, modifying , structuring data, and data warehousing; while they need this time to do their core work of Data Analysis, Data, Mining, ML,DL and “AI” in generaletc.
 - Hence, the effective data preparation techniques will let DS focus more on data mining and data analysis and generate business value.
 - **In the ideal process, Data PREP must be enabled to be done automatically.**





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Values that Data PREP will add....

- Ensure Reliable data for Reliable mining & “AI”.....;
- Detect Data Errors and Issues,
- Provide Better tools for decision-making to the Executive Managers.
- Effective BI process for data analysis.
- Systematic approach to deal with Big Data to combine structured, semi structured and unstructured data, often in raw form until it's needed for specific analytics uses.





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Data PREP Phases:

➤ **Data collection. :**

- It will be gathered from different sources.data scientists, BI team, data professionals and end users who collect data should confirm that it's a good fit for the objectives of the planned BI, analytics applications, ML...."AI"

➤ ***Data Profiling, Scaling, and Understanding:***

- Data needs to be profiled and explored, what's in it and how could we prepare it for the next stages or use
- Data patterns; how these data related to each other, what value might have, is there any errors, missing part, consistency issues..

➤ **Data cleansing :**

- Correction and editing what's missing, consistency maintenance
- faulty data is removed or fixed, missing values are filled in and inconsistent entries are harmonized.





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➤ Data structuring:

- Data should be organized to meet the analytics requirements.
- Data stored in comma-separated values (CSV) files or other file formats has to be converted into tables make it ready for further use in BI, Data analytics, ML, "AI" applications

➤ Data transformation:

- Usable format, creating new fields or columns that aggregate values from existing ones.
- Further enhances and optimizes data sets as needed, through measures such as augmenting and adding data.

➤ Data validation:

- Develop and automated routines that can check data validity, its consistency, completeness and accuracy.





Data Pipelines:

- Typical in the recent days, businesses generates huge volume of data. This data is distributed among different systems used by the business: Cloud Applications, Databases, SDKs, Outsourced Tools ...etc.
- To obtain valuable insight from such a highly distributed data, it should be merged, integrated or even moved to a single location for easy access and automated analysis.
- A Data Pipeline is the mechanism by which ETL processes occur.
- We need to be able to distinguish among: Batch vs Real-Time Data Pipeline, Open-Source vs Proprietary Data Pipeline, and On-premise vs Cloud Data Pipeline.





Data Pipelines: Tools and Techniques

- ETL can be done manually. Alternatively, each of these steps can be automated using separate software tools, "AI" is a good option.
- However, during the process, there are many things that can break.
 - The code can throw errors, data can go missing, incorrect/inconsistent data can be loaded, and so on.
- Often, Data Pipeline tools are used to automate this process end-to-end in an efficient, reliable, and secure manner.
- Data Pipeline software guarantees consistent and effortless migration from various data sources to a destination, often a Data Lake or Data Warehouse.





Data Pipelines: Types, Tools and Techniques

- Depending on the purpose, there are different types of Data Pipeline tools available. The major considerations in selecting specific tool or applying any technique are:
 - Batches
 - Real-Time
 - On-Premise
 - Cloud
 - Open Source
 - Proprietary





Data Pipelines: Types, Tools and Techniques

- Pipelines that handle Batch Data would let the system moving data in a very large amount at regular time frame or regular batches.
- This type of operation could fit more for the on-premise data sources.
- Still Real time data can be collected in an instance of batches, in cases where real-time processing can constrain regular business operation due to limited resources.





Data Pipelines: Types, Tools and Techniques

- ETL tools could be optimized to process the Pipelines actives in real-time.
 - **Therefore, such a tools would be excellent if we are performing some data analysis on the fly.**
- Real time data Pipelines will be extremely useful when we need to process data coming, for example, from our web-based applications or mobile apps to extract some real time insight or to make real time decision or real time diagnosis.
- Apache Kafka is an open-source stream-processing software platform that was initially developed by LinkedIn and became open-sourced in 2011.
 - Kafka has various use cases, but it is primarily used to build real-time streaming data pipelines and applications.





Data Pipelines: Types, Tools and Techniques

- Utilizing open-source data Pipelines tools can be customized for certain needs and applications, expand its features, at the same time this would requires an experience in maintaining the tools.
 - However, the underlying technology of the tool is publicly available.
- On the other hand, we can just utilize proprietary data pipeline tools that could be configured for your business.





Data Pipelines: Types, Tools and Techniques

- In our business we used to store all our data on our local servers.
- In addition, we used to build our data warehouse and data lake locally on-premise as well.
- The data pipelines for this type of data offer better security as they are deployed on our local infrastructure.
- Cloud Data Pipeline tools would let us transfer and process of Cloud data to Data Warehouses hosted in the cloud or even locally, cloud service providers put enough focus on security as well.





Data Warehousing

- “A data warehouse is a **subject oriented, integrated, time-variant, and non-volatile** collection of data”. Inmon.
- Operational database undergoes frequent changes on a daily basis on account of the transactions that take place.
- A data warehouses provides us with a generalized and consolidated data in multidimensional view, and it provides an Online Analytical Processing (OLAP) tools.
- OLAP tools help us in interactive and effective analysis of data in a multidimensional space for Data mining functions such as association, clustering, classification, prediction that can be integrated with OLAP operations to enhance the interactive mining of knowledge at multiple level of abstraction.
 - As such, data warehouse has now become an important platform for data analysis and online analytical processing, and more for data mining, ML,.....





Data Warehousing

- A data warehouse is a database, which is kept separate from the organization's operational database.
- There is no frequent updating done in a data warehouse.
- It has consolidated historical data, which helps the organization to analyze its business.
- A data warehouse helps executives to organize, understand, and use their data to take strategic decisions.
- Data warehouse systems help in the integration of diversity of application systems.
- Subject oriented, Integrated, Time-Variant, Non-Volatile





Data Warehousing

Types of Data Warehouse

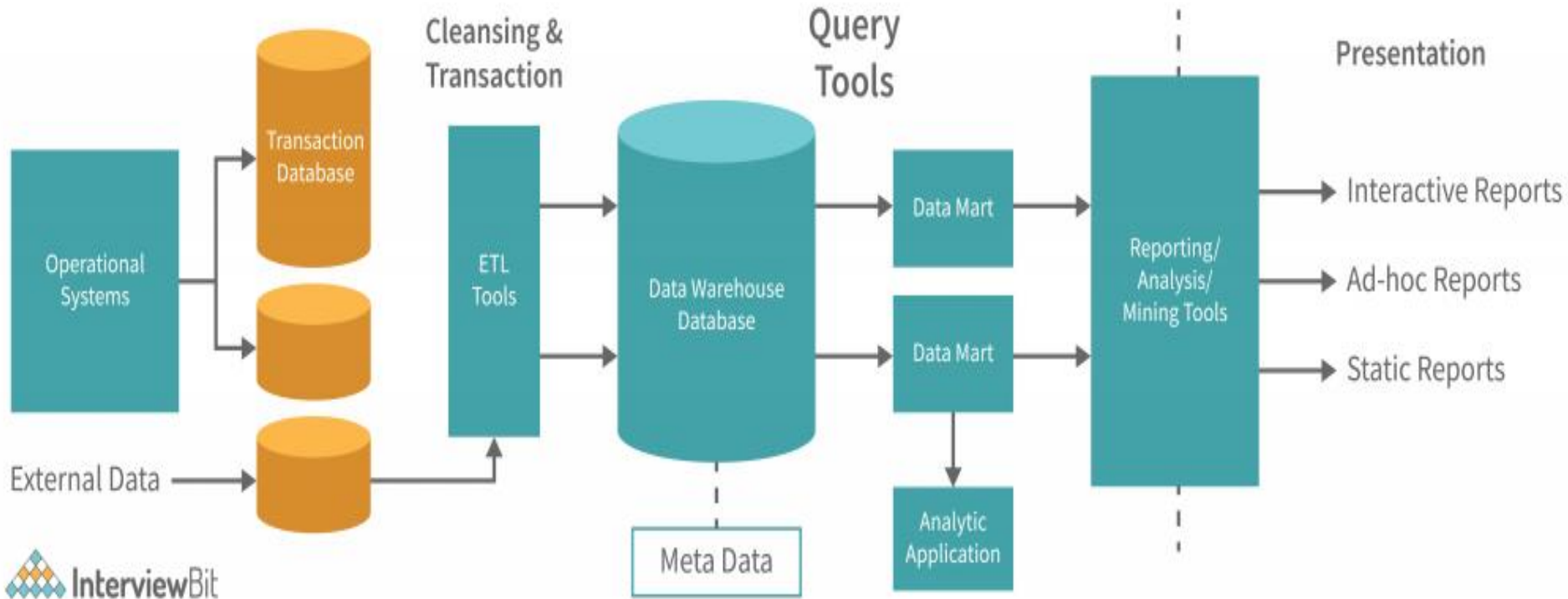
- Information Processing – A data warehouse allows to process the data stored in it.
 - The data can be processed by means of querying, basic statistical analysis, reporting using crosstabs, tables, charts, or graphs.
- Analytical Processing – A data warehouse supports analytical processing of the information stored in it. The data can be analyzed by means of basic OLAP operations,
- Data Mining – Data mining supports knowledge discovery by finding hidden patterns and associations, constructing analytical models, performing classification and prediction.
 - These mining results could be presented using the visualization tools.
 - To integrate heterogeneous databases, two approached
 - Query-driven Approach
 - Update-driven Approach





Data Warehousing

Data Warehouse Architecture



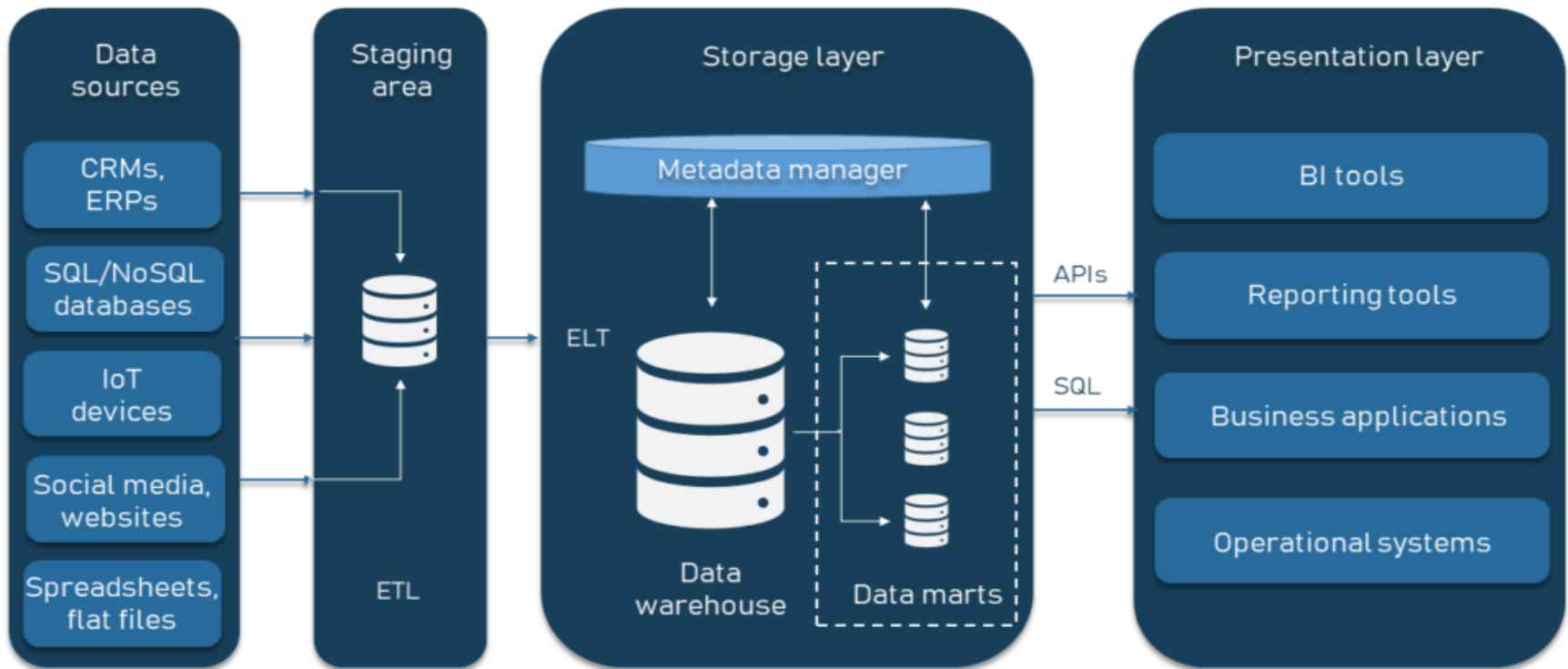
<https://www.interviewbit.com/blog/data-warehouse-architecture/>





Data Warehousing:

ENTERPRISE DATA WAREHOUSE COMPONENTS



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